CEBU INSTITUTE OF TECHNOLOGY - UNIVERSITY
College of Computer Studies, IT411 Capstone & Research 2

Face-to-Face Adviser Consultation Master Briefing & Live Demo Guide

Consultation Agenda: (A) Software Functionality, (B) Deployment & Repo, (E) Testing, QA & Survey Synthesis

Document ID:	PLAYIT-F2F-BRIEFING-v1.0
Course:	IT411 — Capstone & Research 2 | Semester 1, AY 2026–2027
Institution:	College of Computer Studies, Cebu Institute of Technology – University
Capstone Adviser:	Mr. Joemarie C. Amparo
Consultation Agenda:	`A. Software Functionality and Completion | B. Deployment and Source-Code Repository | E. Testing and Quality Assurance (Survey Responses and Synthesis)`
Team Members:	Zendrix (`zendrix-hub`, Team Lead) and Groupmates

1. Quick-Reference Readiness Scorecard for Mr. Amparo's 3 Agenda Items

Adviser Agenda Item	Primary Files to Open During F2F	Headline Verification Proof
A. Software Functionality & Completion	- Physical Android phone running `playit-debug-H-0320385.apk` (or `playit-debug-card32-latest.apk`) - `01_Software_and_Deployment/FEATURE_COMPLETION_MATRIX.pdf` - `01_Software_and_Deployment/CODE_REFACTORING_EVIDENCE.pdf`	- **14 / 14 functional requirements (`FR-01`–`FR-14`) implemented and verified** across 12 Jetpack Compose screens and 12 Room SQLite tables. - **Cards 00–32 completed** on `refactor/hear-say-it` (`9031d8f`): 4-step pure-phoneme `Hear It`, offline Vosk grammar + 3-attempt scaffolding + mouth cue `Say It`, 5-card `Find It`, 33-word deranged `Blend It`, Avatar-only onboarding, and 1-page offline PDF Parent Report. - **Zero `INTERNET` permission** (`100%` offline operation).
B. Deployment & Source-Code Repository	- `01_Software_and_Deployment/DEPLOYED_SYSTEM_AND_ACCESS_INSTRUCTIONS.pdf` - `01_Software_and_Deployment/DEPLOYMENT_AND_PERMISSIONS_INSPECTION.pdf/.md` - Live GitHub: `https://github.com/zendrix-hub/PlayIT` (`6ca3315`) & Draft PR #2 (`https://github.com/zendrix-hub/playIT-v2/pull/2`)	- **Submitted Midterm APK H (`playit-debug-H-0320385.apk`):** `108,317,310` bytes, SHA-256 `b5a54c18d1405c3851b7fec5c110f03f1aeaaa899a343b12f1796e04398c42d4`. - **Clean Architecture Verified:** `domain/` contains **0** `android.*` **imports** (`DomainPurityTest`). - **Asset Release Gates Verified:** 201 audio clips (`Kokoro-82M`, `Chatterbox-Turbo`, `ElevenLabs`) and 80 child-safe WebP illustrations tracked via release manifests.
E. Testing & QA (Survey Responses & Synthesis)	- `03_Testing_and_QA/SURVEY_RESPONSES_AND_SYNTHESIS_REPORT.pdf` - `03_Testing_and_QA/Cleaned_Master_Responses_N25.xlsx` & `ROUND1_SURVEY_RESPONSES_TABULATION_N25.csv` - `03_Testing_and_QA/ACCEPTANCE_TESTING_AND_SIGNOFF_REPORT.pdf` (Signed) - `02_Updated_Documentation/playIT_STD_v1.0.pdf` (Signed) - `04_Final-User_Validation/SMART_Goal_Matrix.pdf` & `Final_Validation_Instrument.pdf`	- **Round 1 Survey (N = 25):** Cohort 1 Children (n = 16, `93.8%` Enjoyment, `56.3%` Perceived Ease), Cohort 2 Parents (n = 5, Mean SUS `75.50`, Grade-Fit `4.60/5.00`), Cohort 3 DepEd Grade 1 Teachers (n = 4, `100%` curriculum alignment, `50%` `PED-08` phoneme critique that drove the Hear It / Say It overhaul). - **Software QA: 433 automated tests** in STD v1.0.1 (`435` on `HEAD`), **83 functional test cases**, **7 closed defects (`INC-01`–`07`)**, **2 transparently tracked open items (`DEF-01`, `DEF-02`)**, and signed physical **Samsung Galaxy A21s** acceptance.

2. Part A — 5-Minute Live Software Transactions Walkthrough Script

Before starting the demo on the Android phone, turn **Airplane Mode ON** (and Wi-Fi OFF) in front of Mr. Amparo to prove visually that PlayIT operates **100% offline** with zero network dependency.

Step & Time	Screen / Feature	Exact Action to Perform on Phone	What to Point Out to Mr. Amparo (Transaction & DB Proof)
Transaction 1 (0:00–0:45)	**Splash & Avatar-Only Onboarding (**FR-01, FR-14**)**	Launch PlayIT → Tap **"New Learner"** on `ProfileSelectScreen` → Pick one of the 6 illustrated avatars → Tap **Start**.	- **Survey Synthesis Link:** In Round 1, Grade 1 children struggled to type their names on an Android keyboard. Card 08 replaced text typing with **1-tap Avatar-Only Onboarding**. - **Database Transaction:** Inserts a new row into Room table `profiles` (`ProfileEntity`) and sets `SessionManager.activeProfileId`.
Transaction 2 (0:45–1:15)	**Curriculum Map & Chapter Gating (**FR-05, FR-08**)**	Show the vertical scrolling Archipelago Map (`CurriculumMapScreen`) with 7 chapters (26 English letters, Marungko sequence starting with Chapter 1 `M, S, A, I`) and 16 Philippine landmarks. Tap a locked node (S), then tap unlocked node `M`.	- **Business Rule Proof:** Locked nodes cannot be entered until the preceding letter is completed (`LessonUnlockManager`). - **Scope Resolution:** Ñ and `NG` were formally removed in Card 14 (`FR-05`) because they lack English phonics exemplars.
Transaction 3 (1:15–2:00)	**Stage 1:** `Hear It` **Stepped Phoneme Modeling (**FR-02, NFR-AUD-01**)**	Enter Letter `M` **(**`Hear It`**)**. Let Lily play the 4-step sequence: Step 1 (`"Listen!"`) → Step 2 (`/m/` played 3 × with 700 ms pauses) → Step 3 (`"mouse"` card reveal) → Step 4 (`/m/` closure). Point out that the **Next** button is disabled until Step 4 finishes.	- **Survey Synthesis Link:** Directly resolves **Teacher Finding #1 (**`T-1` **Leony Layaguin &** `T-3` **Teacher Joy)**: *"The sound of M should be pronounced as /m/ (mmm, mmm, mmm) rather than 'ma, ma, ma.'"* - **Audio Engineering:** Uses voice-matched Chatterbox-Turbo/Kokoro pure continuant `/m/` (`"Mmm!"`) with zero intrusive schwa vowel.
Transaction 4 (2:00–3:15)	**Stage 2:** `Say It` **Offline Vosk Speech Recognition & 3-Attempt Scaffolding Ladder (**FR-03**)**	1. **Attempt 1 (Intentional Foil):** Tap the Mic (`Idle` → `Listening` → `Heard` → `Result`) and say the letter name `"em"` (or `"muh"`). Show Lily's specific corrective feedback (*"That's the letter name! Make just the /m/ sound"*). 2. **Attempt 2 (Mouth Cue):** Point out the `ArticulationCue` (`lips_together` diagram) that appears beside the mic on Attempt 2. 3. **Attempt 3 / Correct Utterance:** Say `"mmm"` (or `"mouse"` in echo step) to pass, OR fail a 3rd time to demonstrate **Echo Together** auto-advance with **zero heart loss**.	- **Survey Synthesis Link:** Directly resolves **Child Ease Friction (**`Q2 = 56.3%`**)** and **Accessibility Flag** `ACC-06` **(**`60%` **hearing support)**. - **Technical Architecture:** `VoskSpeechRecognitionService` runs a constrained JSON grammar `[target, foils..., "[unk]"]` completely offline; `SpeechValidator` classifies utterances into `Correct`, `LetterName`, `AddedVowel`, `OtherWrong`, or `Silence`; Room records every attempt in `say_it_attempts`.
Transaction 5 (3:15–4:00)	**Stage 3:** `Find It` **5-Card Picture Discrimination & Heart/Star Gamification (**FR-04, FR-07, FR-09**)**	On `Find It`, show the 5 illustrated cards (**3 target** `/m/` **pictures** + **2 non-**`/m/` **distractors** from our 80 new child-safe WebP illustrations). Tap 1 wrong card to show **1 heart deducted** (`5 -> 4`), then tap all 3 correct `/m/` cards (`mouse`, `mango`, `milk`) to reach `LetterCompleteScreen` (1–3 stars).	- **Database Transaction:** Logs each card tap into `find_it_attempts`, updates `lesson_progress` (`bestStars = max(existing, new)`), and unlocks the next letter node on the map.
Transaction 6 (4:00–4:30)	**Checkpoint:** `Blend It` **Deranged Word Construction (**FR-13**)**	Open Chapter 1 `Blend It` (`SAM` / `AM`). Show that the letter tiles (S, A, M) are **always dealt in a deranged (unsolved) order** (`TC-BLD-02`). Tap the tiles in order S → A → M to hear the blended word.	- **Curriculum Fix (**`Card 15`**):** Replaced the 5 non-decodable sight words (`AIM, BEE, TOY, BOY, ZOO`) with 100% Marungko-decodable words (`AM, SUM, TUB, YAM, ZIP`) across the 33-word bank.
Transaction 7 (4:30–5:00)	**Parent Zone Arithmetic Gate & Offline PDF Progress Report (**FR-10, FR-11, FR-12**)**	Tap the Parent Zone icon → Solve the **two-digit addition gate** (e.g., `24 + 17 = 41`) → Show per-letter mastery, speech accuracy, ElevenLabs attribution, and tap **Export PDF Report** (`ReportPreviewScreen`).	- **Technical Proof:** Uses Android's native `android.graphics.pdf.PdfDocument` (`PdfExporter.kt`) to render a clean, 1-page clinical progress summary completely offline (`report_logs` table).

3. Part B — Deployment & Source-Code Repository Reference Sheet

3.1 Live URLs, Branches & Binary Checksums

Item	Verified Value
Public Midterm Snapshot Repository	`https://github.com/zendrix-hub/PlayIT` (Branch `main`, tag `midterm-submission-2026-10`, commit `6ca3315`)
Active Development Repository & PR	`https://github.com/zendrix-hub/playIT-v2/pull/2` (Branch `refactor/hear-say-it`, Cards 00–32)
Frozen Midterm Binary (`APK H`)	`playit-debug-H-0320385.apk` (`108,317,310 bytes`, commit `0320385`) **SHA-256:** `b5a54c18d1405c3851b7fec5c110f03f1aeaaa899a343b12f1796e04398c42d4`
Post-Card-32 Latest Demo Binary	`playit-debug-card32-latest.apk` (Built from `refactor/hear-say-it` HEAD `9031d8f`, includes Card 26c 80 WebP illustrations + Card 32 palette/PDF polish)
Midterm Submission Google Drive	`https://drive.google.com/drive/folders/1093ZTtHqgy6k-3D2V8FPYxK_LOiu8iL7`
Pre-Midterm Survey Evidence Drive	`https://drive.google.com/drive/folders/1JhVlGw1ncpvOF6lQjBOQC4gwXKZdMbib`
Zero-Internet Offline Proof	Verified via `aapt dump permissions playit-debug-H-0320385.apk`: requests **only** `android.permission.RECORD_AUDIO` and `android.permission.VIBRATE` (**zero** `android.permission.INTERNET`).

3.2 Honest Engineering Disclosure on Open Defects & Deferred Scope (`STD v1.0.1` & SRS `v3.0.1`)

If Mr. Amparo asks *"Are there any known limitations or open bugs in your submitted midterm build?"*, give this exact, transparent engineering answer:

1. `DEF-01` **(Medium — Tracked in** `DEFECT_LOG_AND_RETEST_RECORD.pdf`**):** In `FindItViewModel.kt`, the 2 distractor cards are currently drawn from a static fallback list of 3 exemplars (`s_sun`, `a_apple`, `t_turtle`) instead of dynamically querying `PhonemeDao.getByChapter()` for other letters. **Impact:** Gameplay works without crash (5 cards: 3 targets + 2 distractors), and dynamic distractor selection is scheduled as a post-validation improvement.
2. `DEF-02` **(Low — Tracked in** `DEFECT_LOG_AND_RETEST_RECORD.pdf`**):** `StarCalculator.calculate()` currently computes stars from `heartsRemaining` (`5 hearts = 3 stars`, `3–4 hearts = 2 stars`, `1–2 hearts = 1 star`), while the combined accuracy term in SRS `FR-09` is documented for alignment after the Week 12 empirical validation.
3. **Planned Post-Midterm Features (**`FR-NEW-TEL` **&** `FR-NEW-REC`**):** `SayItAttemptEntity` already logs `attemptNumber`, `recognizedText`, `isPassed`, and `timestamp` in Room SQLite (`export_validation_telemetry.py` extracts this for Week 12 validation). Adding the `errorType` column (`FR-NEW-TEL`) and the pure-phoneme recall prompt (`FR-NEW-REC`) are documented as post-midterm enhancements.

4. Part E — Survey Responses, QA Synthesis & Final Validation Gate

4.1 Summary of Round 1 Survey (N = 25) vs. Round 2 Final Validation Plan (N = 28–37)

Metric / Dimension	Round 1 MVP Survey Result (Sept 16–23, 2026, N = 25)	Refactored Software Fix (Cards 00–32)	Round 2 Final Validation Target (Due Oct 31, 2026, N = 28–37)
Child Perceived Ease & Task Completion	Smileyometer `Q2` Ease = `56.3%` **Top-Box** (9/16); `Q1` Fun = `93.8%` (15/16); `Q3` Characters = `93.8%` (15/16); `Q4` Replay = `87.5%` (14/16).	Avatar-only onboarding (`Card 08`), 4-state Mic (`Card 06`), 3-attempt scaffolding ladder (`Card 05`), `ArticulationCue` mouth diagram (`Card 16`).	**SMART Goal** `SG-01`**:** Task completion rate **≥ 85%** across Scenarios S1–S4 (`Instrument A` + `Instrument C`), Child Positive Affect **≥ 85%**.
Phoneme Audio Accuracy (`PED-08`**)**	DepEd Teachers rated `PED-08` `50.0%` **(**`2/4`**)**: `T-1` (L. Layaguin) & `T-3` (J. Flores) flagged `/m/` sounding like `"ma, ma, ma"` or letter name. All other 11 items = `100%` **(**`4/4`**)**.	Re-synthesized 201 audio clips (`Kokoro-82M`, `Chatterbox-Turbo` `"Mmm!"`, `ElevenLabs` short vowels/blend words) + 4-step `Hear It` sequence (`Card 04`).	**SMART Goal** `SG-02`**: 100% (**`26/26`**)** phoneme clip pass rate on `Instrument B (Part B1)` Teacher Phonetic Audit (`Gate 3`).
Offline Speech Recognition (`Say It`**)**	Unconstrained Vosk accepted foils or caused frustration; hearts were deducted on speech errors.	Constrained Vosk grammar `[target, foils..., "[unk]"]` (`Cards 01–03`), `SpeechValidator` classification, **zero heart loss in Say It**.	**SMART Goal** `SG-03`**:** Teacher-vs-Vosk agreement **≥ 80%** and false-rejection rate **≤ 15%** across ≥ 60 utterances (`Instrument B Part B2`).
Curriculum & Pedagogical Validity	`100%` (`4/4`) on binary checklist (`PED-01..07, 09..12`), but lacked formal Content Validity Index (`CVI`) and had 5 sight words in `Blend It`.	26-letter Marungko curriculum (`Card 14`), 33 decodable CVC/VC words (`Card 15`), deranged tile dealing (`Card 11`), 80 child-safe WebP pictures (`Card 26c`).	**SMART Goal** `SG-05`**:** Lynn (1986) `I-CVI` **≥ 0.83**, `S-CVI/Ave` **≥ 0.90**, and TAM `PU` **&** `PEOU` **≥ 4.00/5.00** (`Instrument B Part B3`).

Metric / Dimension	Round 1 MVP Survey Result (Sept 16–23, 2026, N = 25)	Refactored Software Fix (Cards 00–32)	Round 2 Final Validation Target (Due Oct 31, 2026, N = 28–37)
Parent Usability & Report Clarity	Mean SUS = 75.50 (n = 5, SD = 14.62, Grade B; P-2 = 50.0 acquiescence); Grade-Fit = 4.60/5.00; ACC-06 Hearing = 60%.	Added ArticulationCue visual mouth diagrams (Card 16), 1-page offline PDF report (Card 32 clinical palette), and ElevenLabs attribution (Card 15b).	**SMART Goal** SG-06**: Mean Parent SUS ≥ 75.0** across expanded **n = 10–12 parents** (Instrument E) and Report Clarity **≥ 4.00/5.00**.

4.2 Pre-Presentation Deliverables Ready for Mr. Amparo's Sign-Off (04_Final-User_Validation/)

To secure Mr. Amparo's approval to proceed with Week 12 Final User Validation (due October 31, 2026), present these four completed documents inside 04_Final-User_Validation/:

1. SMART_Goal_Matrix.pdf / .docx — Maps all 6 measurable SMART goals (SG-01 to SG-06) to their exact indicator, instrument item, formula, and acceptance threshold.
2. Approved_Proposal_and_SMART_Goals.pdf / .docx — Preserves the original Capstone 1 proposal baselines and traces every adviser-approved scope refinement.
3. Final_Validation_Plan_and_Framework.pdf / .docx — Defines the 5 research questions (RQ1–RQ5), the 3 stakeholder cohorts (N = 28–37), and the triangulated evaluation framework (ISO/IEC 25010 + Davis TAM + Brooke SUS + Lynn CVI + Read & MacFarlane Smileyometer).
4. Final_Validation_Instrument.pdf / .docx — Contains the 5 ready-to-print evaluation instruments (Instruments A, B, C, D, and E).